

NETWORKS AND COMPLEXITY

Solution 12-10

*This is an example solution from the forthcoming book Networks and Complexity.
Find more exercises at <https://github.com/NC-Book/NCB>*

Ex 12.10: Change of variables [Lvl. 3]

Consider a system of differential equations where

$$\begin{aligned}\dot{x} &= y \\ \dot{y} &= -x\end{aligned}$$

Transform these equations into polar coordinates

$$\begin{aligned}r &= \sqrt{x^2 + y^2} \\ \phi &= \arctan(y/x).\end{aligned}$$

Then solve the differential equations.

Solution

Computing the time derivative of r we find

$$\dot{r} = \frac{1}{2} \frac{1}{\sqrt{x^2 + y^2}} (2x\dot{x} + 2y\dot{y}) \quad (1)$$

$$= \frac{x\dot{x} + y\dot{y}}{\sqrt{x^2 + y^2}} \quad (2)$$

Now substituting the equations for x and y we find

$$\dot{r} = \frac{xy - yx}{\sqrt{x^2 + y^2}} = 0 \quad (3)$$

That is nice. Now consider ϕ .

$$\dot{\phi} = \frac{d}{dt} \arctan(y/x) \quad (4)$$

$$= \frac{1}{1 + y^2/x^2} \left(\frac{d}{dt} \frac{y}{x} \right) \quad (5)$$

$$= \frac{1}{1 + y^2/x^2} \left(\frac{\dot{y}}{x} - \frac{\dot{x}y}{x^2} \right) \quad (6)$$

$$= \frac{1}{1 + y^2/x^2} \left(-1 - \frac{y^2}{x^2} \right) \quad (7)$$

$$= -\frac{1 + y^2/x^2}{1 + y^2/x^2} \quad (8)$$

$$= -1 \quad (9)$$

So it turns out that r is actually a conserved quantity. It is thus a parameter of our system and not a variable. We only need to integrate the equation for ϕ , which we can do by direct integration

$$\phi(t) = \int -1 dt = -t + C. \quad (10)$$

To translate back into our original coordinates we need the inverse coordinate transformation, which is

$$x = r \cos \phi \quad (11)$$

$$y = r \sin \phi. \quad (12)$$

Substituting our solution for ϕ yields

$$x(t) = r \cos(C - t) \quad (13)$$

$$y(t) = r \sin(C - t) \quad (14)$$

so the system is going in a circle. We have found a periodic solution.